LFRViT: Laplacian Filter Residual-based Vision Transformer for Multiple Image Forgery Detection

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***Abstract*— When producing a manipulative image, a forger can alter an image through a variety of image tampering techniques. Due to the need to test for various image editing operations and alterations, there has been a significant interest in developing a universal image forgery detection approach that can detect multiple tampering operations performed over an image. This paper presents a comprehensive forensic method for detecting manipulation using Vision Transformer. We present a novel network architecture called Laplacian Filter Residual-based Vision Transformer (LFRViT) that can automatically learn features for detecting manipulation from training data. Vision Transformers, as they are now designed, primarily to learn features that represent the content of an image rather than features that detect manipulation. To address this problem, we have created a novel type of model specifically designed to suppress the image's content and dynamically acquire features for detecting manipulations. By conducting a sequence of studies, we provide evidence that our proposed method can automatically learn the features to identify various image alterations. The experimental findings demonstrate that our proposed model, LFRViT, is capable of autonomously identifying various types of manipulations with an accuracy of over 99%.**

Keywords—Vision Transformer, Laplace, Image Forgery Detection, Deep Learning, Image Forensics

I. Introduction

In the digital age, as images are used extensively in documentation, entertainment, and communication, it is critical to assure their authenticity. However, image manipulation and forgery have become more sophisticated and accessible with the development of image editing software [1]. This has sparked questions about how dependable and credible the visual content that is being shared online and in other media is.

The field of research and technology known as "multiple image forgery detection" is devoted to detecting situations in which multiple images are combined or altered to produce an inaccurate or misleading representation. Multiple-image forgery detection exposes instances of manipulation or tampering by examining relationships and inconsistencies between multiple images, in contrast to traditional single-image forgery detection, which focuses on finding a single tampering operation within an image [1].

Effective techniques to identify manipulated or tampered images are more important than ever due to the spread of social media sites, online news sources, and digital archives. These kinds of images can be used to disseminate false information, sway public opinion, or trick people or institutions. Thus, in the larger field of digital forensics and image analysis, the creation of efficient methods and algorithms for multiple image forgery detection has become essential research. Through the application of advances in machine learning, computer vision, and signal processing, scholars and practitioners aim to improve the veracity and authenticity of visual content on digital platforms [2].

In this study, we have considered four different types of image tampering operations. These tampering operations are applied to each original image to have a manipulated image. The four different types of tampering operations are: Additive White Gaussian Noise (AWGN), resampling, median filtering, and gaussian blurring. Fig. 1, shows the different tampering operations performed over the original image, taken from the RAISE dataset.

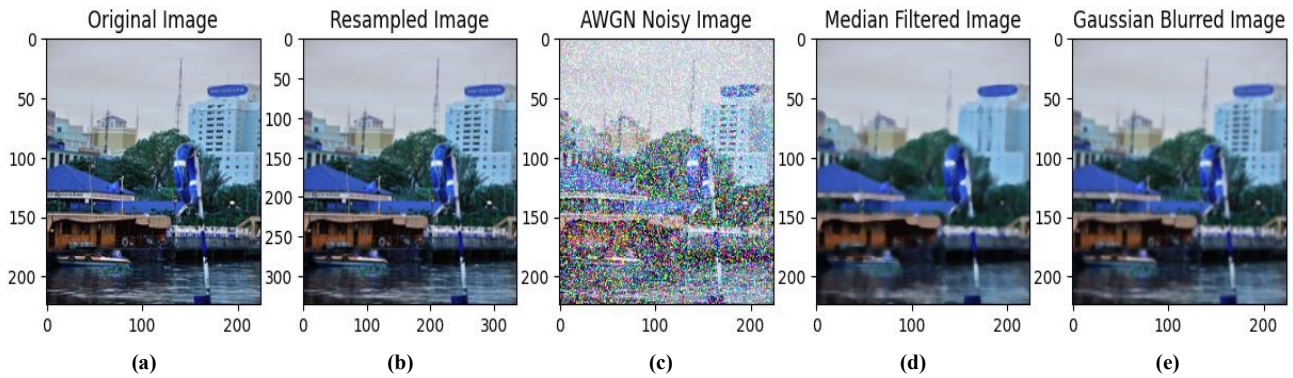

Fig.1. From RAISE database, **(a)** original image and different operations are performed on this original image, **(b)** resampled image with a scaling factor of 1.5, **(c)** AWGN noisy image with standard deviation of 2, **(d)** median filtered image with a 5 × 5 kernel size and **(e)** Gaussian blurred image with 5 × 5 kernel and σ = 1.1.

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This study introduces a novel universal method for detecting image editing, which has the ability to autonomously acquire knowledge about the traces left by the editing operation. In order to achieve this, we utilize the Vision Transformer (ViT) [3]. ViTs have recently led to significant progress in image recognition by being able to dynamically learn classification features instead of relying on hand-crafted features. We present a deep-learning approach that turns local image regions into masked features using patch-level learnable masking. The ViT model receives these masked features in order to identify any global discrepancies between the local masked features that are produced. These generated features are highly valuable in the manipulation detection area because they can efficiently learn the residual artifacts in manipulated images, whether they are local or global. From now on, we will refer to this model as Vision Transformer with Laplacian filter residual layer, or in short LFRViT. Fig. 3, shows the entire architecture of our proposed model LFRViT.

II. Related Work

In recent years, researchers have created various information forensic methods to ascertain the genuineness and processing history of digital images [1], [4]. A significant portion of this research has concentrated on the identification of distinct traces or clues that are left behind in an image as a result of particular editing actions. Subsequently, algorithms have been devised with the specific purpose of detecting these marks or indications. This methodology has been employed to create algorithms specifically designed to identify image alterations, such as changes in size and resolution [5], [6], application of median filtering [7]–[9], enhancement of contrast [10], and so on.

Although the progress in developing targeted editing detectors has resulted in significant advancements in information forensics, this approach to image authentication has a notable limitation: Given the wide range of editing techniques available to a forger, a forensic investigator must employ numerous forensic tests to ascertain the presence and extent of image manipulation. When multiple forensic tests are conducted on an image, the investigator must tackle various new challenges, including managing the overall false alarm rate across multiple tests and resolving conflicting results. In addition, researchers must identify the residual effects of newly developed image editing operations and create corresponding detection algorithms.

Given these concerns, there has been considerable attention focused on creating universal forensic algorithms that can identify a wide range of editing operations, if not all of them. Recent empirical evidence has demonstrated that tools originally designed for steganalysis have the ability to identify a diverse range of image editing operations [11], [12]. These steganalysis tools function by constructing local models of pixel interdependencies through the examination of the combined distribution of prediction errors in pixel values. Detection features are subsequently derived from these combined distributions [13]. A recent endeavor in the field of forensic detection involves the creation of Gaussian mixture models (GMMs) using image patches from both unaltered and manipulated images [14]. Multiple binary manipulation detectors are employed to compare the log-likelihood of an image patch under the Gaussian Mixture Model (GMM) for various editing operations. Although these techniques exhibit significant potential, they acquire detection features from pre-determined models. Consequently, a pertinent inquiry arises: Is it possible to identify robust universal detection characteristics without the need for human examination or the imposition of a pre-established model on the data?. Also, various spatial domain and frequency domain techniques have been created to detect these spatial-frequency alterations [15].

The subsequent sections of this paper are organized as follows. Section 3 contains specific information regarding the proposed method, whereas Section 4 provides detailed information about the experimental setup. The results are presented and analyzed in Section 5. The paper is concluded in Section 6, summarizing the work conducted. The proposed method combines image processing knowledge with a deep learning technique. Initially, we formulated a convolutional layer having a Laplace filter to have residuals and subsequently conducted training on a ViT model for the purpose of classification.

III. PROPOSED METHODOLOGY

This study proposes an LFRViT, a deep learning-based classification model that distinguishes between real and altered images produced by various tampering operations. In our proposed approach, we simply investigate the tampering activities to discern between real and altered images,

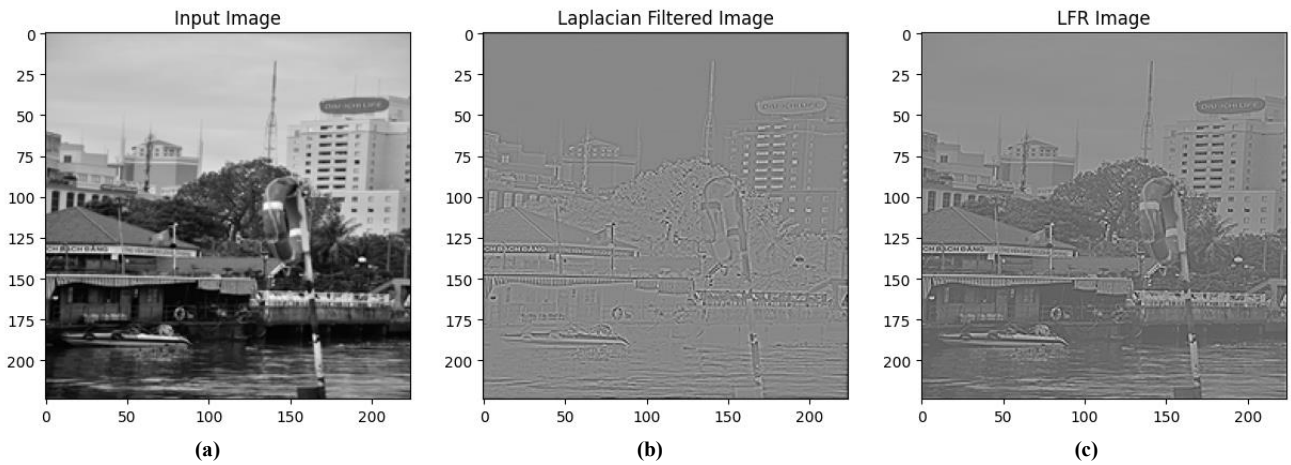

Fig.2. Illustration of the Laplacian filter-based CNN layer output, **(a)** input image, **(b)** Laplacian filtered image obtainrd via (2) and **(c)** LFR image obtained via (3).

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suppressing the image content using the Laplacian filter residual (LFR) [16]. The residual serves as an input for the ViT architecture, which uses it to detect and classify altered images based on their traces. Fig. 2, shows the LFR image corresponding to the input image. The image undergoes some pre-processing before being fed directly as input to the ViT. In pre-processing first, we resize the images to 224 × 224. A Laplacian filter mask of 3 × 3 is utilized, as indicated by (1).

$$Laplacian\ filter\ mask = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (1)$$

The Laplacian of the image is further obtained via (2).

$$L_{i,j} = Laplacian(X_{i,j}) \quad (2)$$

Where $X_{i,j}$ represents the intensity of the pixel at the ith and j^{th} pixels. Finally, subtracting the image from the Laplacian of the image as provided by (3), yields the LFR. As stated, the LFR is given by:

$$LFR_{i,j} = L_{i,j} - X_{i,j} \quad (3)$$

We address the solution to this problem by leveraging the inconsistent occurrence of nearly undetectable residual artifacts in altered images. We have developed a model that is capable of detecting the presence of these artifacts by identifying the inconsistencies within the manipulated image. To do this, we initially apply the Laplacian filter to the input image, resulting in the LFR image. Subsequently, we utilize this LFR image as the input for the ViT model. The ViT model initially divides the image into patches of a predetermined size. Here, we have taken the patch size to be 16 × 16. Every patch is considered a "token," similar to how words are processed in tasks involving natural language processing (NLP). Each patch is encoded into a vector representation using an embedding layer known as token embeddings. The embeddings preserve spatial information of the patches. Positional encodings are incorporated into the token embeddings of the Vision Transformer to compensate for its lack of innate understanding of spatial relationships between patches, an ability that convolutional neural networks (CNNs) possess through convolutions. These encodings convey information about the position of each patch inside the image. The token embeddings, in addition to positional encodings, are subsequently fed into a Transformer encoder. This encoder comprises several layers of self-attention mechanisms, which are then followed by feedforward neural networks. The self-attention mechanism enables the model to capture interdependencies among various patches in the image. Ultimately, the result of the Transformer encoder is sent into a classification head, which usually comprises one or more fully connected layers. This classification head generates the final output probabilities for various classes [3]. Fig. 3, provides a visual representation of the operational concept of the proposed model LFRViT.

IV. EXPERIMENT

This section presents comprehensive information about the datasets utilized for experimentation. Subsequently, we describe the process of preparing the train and test datasets. Additionally, we examine the model using various evaluation measures to assess its performance. These setups assist in validating the effectiveness and generalizability of the proposed LFRViT model. The model is implemented in Python programming language using the TensorFlow framework. All of the experiments were conducted on the Jupyter Notebook platform and trained on an Nvidia Tesla K80 GPU.

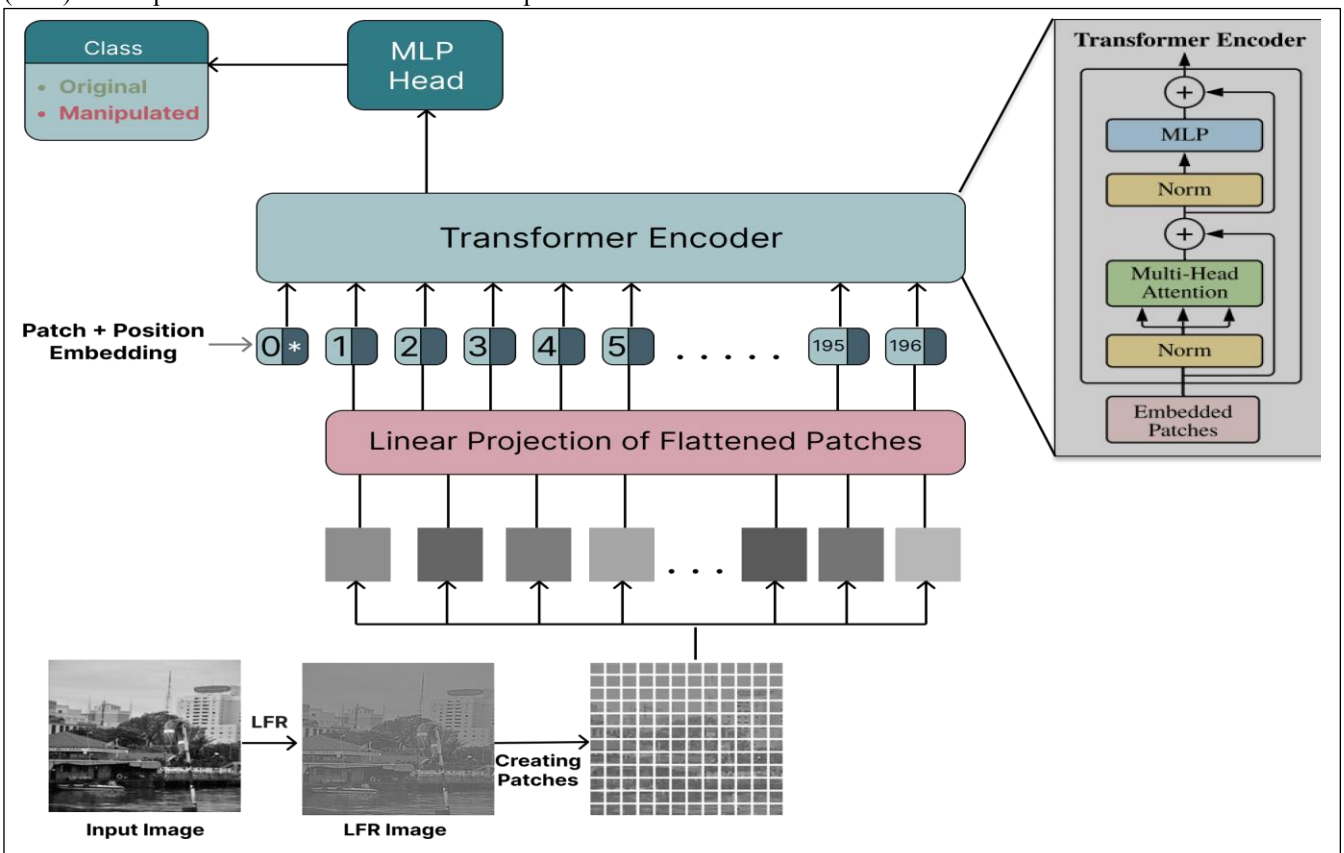

Fig.3. Architecture of the proposed LFRViT model.

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We conducted a comprehensive set of experiments to evaluate the effectiveness of the proposed method in identifying different image-manipulating activities. Standard image datasets, such as Uncompressed Color Image Datasets (UCID) [17], Break Our Stenographic System (BOSSBase) [18], raw images dataset for digital image forensics (RAISE) [19], and the Dresden image dataset (DID) [20], were utilized to create different training and testing sets for different experiments. A total of 33593 images were assembled using 1388 images from UCID with dimensions of 512 × 386, 9074 images from BOSSBase with dimensions of 512 × 512, 15025 images from DID with varying dimensions, and 8156 images from the RAISE dataset. Several training and testing datasets were then created using the original image set as a foundation. Four distinct types of manipulation operations are applied to each image: Additive White Gaussian Noise (AWGN), resampling, median filtering, and gaussian blurring. We then produced a set of altered images. To achieve this, we subjected a collection of unaltered images to the individual manipulation operations such as AWGN with standard deviation of 2, resampling using a scaling factor of 1.5, median filtering with a 5 × 5 kernel and gaussian blurring with a 5 × 5 kernel and a standard deviation of $\sigma = 1.1$.

V. Result

In this section, we present the results of our proposed method to perform both binary and multi-class classification for manipulation detection on an image. Different evaluation metrics are used to assess the model's accuracy in identifying manipulated images. The study utilizes commonly used criteria, including accuracy, precision, recall, and F1 Score, to quantify the effectiveness of classifying manipulated images.

A. Binary Classification

In the initial set of our experiments, we trained our proposed model separately for each of the four manipulations to detect them individually. With the identical architecture described in Section 3, each model corresponds to a binary classifier that can identify a single kind of potential image processing. The original and manipulated images correspond to the two neurons that comprise the output layer. Decision is made by selecting the class corresponding to the highest activated neuron. We constructed our training set for each type of forgery using 26875 unaltered photos and the corresponding manipulated images. Similarly, to create our testing data for each form of forgery, we selected 6718 original images and the corresponding manipulated ones. In other words, we have a total of 53750 training images and 13436 testing images for each binary classification.

The performance of our proposed model for binary classification to identify the underlying manipulating operations is compiled in Table I. This table shows that our proposed approach can differentiate between original and manipulated images with a minimum of 99.32% accuracy, a minimum value of 0.96 for both precision and recall and a minimum value of 0.95 for F1-Score.

B. Multi-class Classification

In the second set of our experiment, our model was trained for multiclass classification to identify several forms of image manipulation, such as median filtering, Gaussian blurring, AWGN, and resampling in comparison to real images. As in the first set of experiments, a choice is made by selecting the class corresponding to the neuron with the highest activation level. We selected 26875 unaltered images and their four corresponding tampered images to create our training set. Similarly, to create our testing data, we selected 6718 original images and the four corresponding tampered images. That is, we have 33590 images for testing and a total of 134375 images for training.

A summary of our simulation results is shown in Table II. The model we propose detects the four main types of forgeries with a minimum accuracy of 99.28%. This confusion matrix shows us how well our model is able to identify each alteration.

There are multiple reasons why these results are significant. First, they demonstrate how our model, which can be trained to identify multiple manipulations without changing its architecture, represents a universal approach to manipulation detection. Perhaps most surprisingly, our model can be taught to automatically learn detection features for every manipulation without the need for human assistance. This implies that our model may be able to learn to detect new manipulations as they are considered or created, eliminating the requirement for a human expert to define detection features.

TABLE I. LFRViT performance as a binary classifier

Evaluation Parameters	Tampering operations			
	Resampling	***AWGN***	***Median Filtering***	***Gaussian Blurring***
Accuracy	99.78%	99.62%	99.32%	99.47%
Precision	0.99	0.98	0.96	0.97
Recall	0.98	0.99	0.94	0.96
F1-Score	0.98	0.98	0.95	0.95

TABLE II. Confusion matrix shows the detection accuracy of LFRViT as a multi-class classifier

	Original	***Resampling***	***AWGN***	***Median Filtering***	***Gaussian Blurring***
Original	**99.61**	1.37	0.51	1.70	1.52
Resampling	1.20	**99.78**	0.18	0.57	0.68
AWGN	0.07	0.09	**99.62**	1.29	1.01
Median Filtering	0.26	0.31	0.51	**99.28**	1.59
Gaussian Blurring	0.72	0.83	0.16	1.30	**99.47**

VI. Conclusion

This work introduces a novel method for universal forgery detection using a Vision Transformer. The method is capable of automatically learning to recognize various types of image manipulations. In order to prevent the Vision Transformer from acquiring knowledge about the visual content of an image, we have introduced a novel type of convolutional layer that utilizes a Laplace filter mask. This mask generates Laplacian filter residuals that are specifically designed to suppress the image's content and facilitate the

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learning of manipulation detection characteristics. We achieved this by deliberately restricting the new convolutional layer to have a Laplace filter. This filter learns the residuals that remain after the tampering process. We have demonstrated, through a sequence of experiments, that our LFRViT model, which is a vision transformer-based universal forensic approach, is capable of automatically acquiring the ability to identify various forms of image manipulations. The experiments conclusively showed that our proposed method can accurately identify various manipulations with an average accuracy of over 99%.

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